

Board Gender Diversity on Firm Performance in the UK: Evidence of Non-linear Returns and Firm Size Moderation (2008–2016)

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ABSTRACT

This paper search for the impact of gender diversity of board of directors on firm performance in the United Kingdom within the period 2008–2016. Drawing on a panel of 840 UK-listed firms, we employ both accounting-based (Return on Assets) and market-based (Tobin's Q) performance measures to test three interrelated hypotheses: (1) a linear positive correlation between the proportion of female directors and firm performance; (2) non-linear effects—whether performance firstly gains rise and then diminish beyond an optimal share of female board members; and (3) the interaction impact of firm size on the diversity – performance link. Our empirical strategy uses ordinary least squares regressions with board size, year, and industry fixed effects, robust standard errors, quadratic terms, and interaction terms. We find no significant linear effect of gender diversity on either ROA or Tobin's Q. However, Tobin's Q exhibits a concave relationship, peaking at approximately 25% female representation, and then declines. Additionally, the positive effects of diversity are significantly larger in smaller firms. These results underscore the importance of tailored diversity targets and firm-specific governance approaches, and suggest that diversity policy should consider non-linear returns and organizational context.

KEYWORDS

Female director; Return on asset; Tobin's Q; Firm organization

1 Introduction

Gender diversity on corporate boards is increasingly regarded as a cornerstone of effective governance. Regulators and industry bodies around the world have paid attention to board-level gender quotas to promote female representation, arguing that diverse decision-making teams enhance oversight and strategic outcomes. Yet empirical studies yield mixed results: while some document argued that female directors enhance firm outcomes, others find negligible or context-dependent effects.

This paper examines the impact of board gender diversity on corporate profitability and market valuation. We analyze both accounting-based and market-based performance—Return on Assets (ROA) and Tobin's Q—using a hand-collected panel of 840 UK-listed firms observed from 2008 to 2016. By situating the analysis in the UK—where several governance reforms and quota initiatives unfolded during this period—we aim to clarify under what conditions and to what extent female representation on boards contributes to firm value.

The following paper is firstly organized Section 2, which reviews the key theoretical arguments and empirical evidence on board gender diversity. Section 3 describes the data structure and variable descriptions. Section 4 shows our main results on linear, non-linear, and size-interaction effects. Section 5 discusses the implications and limitations, and Section 6 concludes with policy implications and suggestions for subsequent work.

2 Literature review

The relationship between gender diversity of board of directors and firm performance has attracted extensive academic investigation, yet remains empirically unsettled. Early cross-sectional studies in the United States documented a positive effect of female directors. Carter, Simkins and Simpson (2003) find that each additional percentage point in board female representation raises Tobin's Q by approximately 0.02, controlling for firm size, industry, and governance variables. Their results offer foundational evidence that diversity may signal superior oversight and stakeholder legitimacy.

Complementing this, Campbell and Minguez-Vera (2008) utilize exogenous gender quota shocks in Spanish firms to establish causality: increases in female director share led to statistically significant gains in Tobin's Q, while reverse causation is ruled out through instrumental-variable tests. This quasi-experimental approach strengthens the claim that gender diversity drives market valuation, rather than merely reflecting previous performance.

Beyond market indications, Adams and Ferreira (2008) delve into the governance mechanisms underpinning diversity effects. Using U.S. board-meeting and committee data, they show that female directors exhibit higher attendance and greater participation in audit and compensation committees, enhancing monitoring intensity. However, in firms with strong governance, excessive oversight may incur coordination costs that actually diminish firm value. These findings motivate the research of non-linear diversity effects in our study, as initial benefits may decrease beyond an optimal

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threshold. Meta-analytic evidence by Post and Byron (2015) provides further nuance. Synthesizing over 140 studies across 35 countries, they report that positive diversity – performance associations are more pronounced in jurisdictions with robust shareholder protections and high societal gender parity. Their analysis also emphasize a stronger link between diversity and accounting-based returns (e.g., ROA) than with market-based measures in some contexts. This suggests institutional factors condition the diversity payoff, justifying our focus on the UK’s well-developed governance framework and the inclusion of year dummies to capture regulatory changes. Finally, Terjesen, Sealy, and Singh (2009) offer a comprehensive theoretical synthesis, identifying four principal lenses—human capital, social capital, resource dependency and institutional theories—through which board diversity can influence firm outcomes. They highlight a persistent gap in empirical research on cross-level mechanisms, particularly how organizational attributes such as firm size moderate diversity effects. Building on this agenda, our paper tests both quadratic diversity specifications and size interactions to illuminate the contextual boundaries of gender board effects.

3 Data and methodology

The dataset consists of 840 observations from UK-listed companies across five biennial periods (2008 – 2016). Key variables include female director proportion (*pfdir*), number of directors (*ndir*), firm size (*lnsize*), and profitability measures (ROA, Tobin’s Q). Econometric models test linear, quadratic, and interaction effects.

Descriptive statistic

Table 1 Summary of the data set

VARIABLES	Description	N	mean	sd	min	max
<i>year</i>	Year in which company observed	840	2,012	2.837	2,008	2,016
<i>roa</i>	Return on Assets = net income as a percent of total assets	840	5.969	12.05	-69.14	43.98
<i>size</i>	Total asset value in millions of GBP	840	1,040	1,481	5.100	12,212
<i>tobinsq</i>	Tobin's Q = total market value / total asset value	840	1.102	1.006	0.0170	7.379
<i>ndir</i>	Number of company directors	840	8.556	2.402	3	19
<i>nfdir</i>	Number of female company directors	840	1.240	1.095	0	6
<i>y10</i>	=1 if observation made in 2010, =0 otherwise	840	0.189	0.392	0	1
<i>y12</i>	=1 if observation made in 2012, =0 otherwise	840	0.193	0.395	0	1
<i>y14</i>	=1 if observation made in 2014, =0 otherwise	840	0.211	0.408	0	1
<i>y16</i>	=1 if observation made in 2016, =0 otherwise	840	0.232	0.422	0	1
<i>industry</i>	Code indicating the industry to which the company belongs	840	52,412	19,404	6,100	93,210
<i>pfdir</i>	Share of females in the directorate	840	0.141	0.119	0	0.545
<i>lnsize</i>	Natural log of size	840	6.277	1.200	1.629	9.410

4 Empirical results

4.1 Linear Relationship between profitability and the female share in a directorate

Initial regressions reveal no statistically significant linear relationship between female director proportion and ROA or Tobin’s Q at the 5% significance level.

For Return on Assets:

$$roa_i = \gamma_0 + \gamma_1 pfdir_i + \gamma_2 ndir_i + \gamma_3 \ln(size)_i + \gamma_4 y10_i + \gamma_5 y12_i + \gamma_6 y14_i + \gamma_7 y16_i + u_{ri}$$

(Model 1_{ROA})

For Tobin’s Q ratio:

$$tobinsq_i = \beta_0 + \beta_1 pfdir_i + \beta_2 ndir_i + \beta_3 \ln(size)_i + \beta_4 y10_i + \beta_5 y12_i + \beta_6 y14_i + \beta_7 y16_i + u_{ti}$$

(Model 1_{TQ})

The share of female directors (*pfdir*) is not statistically significant in predicting either Return on Assets (ROA) or Tobin’s Q when included linearly alongside controls for board size, firm size, year and industry.

By contrast, larger board size (*ndir*) exerts a robust positive effect on both ROA and Tobin’s Q, while greater firm size (*lnsize*) consistently depresses these performance indicators. Year dummies confirm that macroeconomic and industry-wide improvements over time lift both measures.

4.2 Non-linear Relationship between profitability and the female share in a directorate

Quadratic regression results indicate a significant non-linear relationship for Tobin’s Q, suggesting diminishing returns of gender diversity after an optimal level. However, no significant quadratic effect is observed for ROA.

$$roa_i = \gamma_0 + \gamma_1 pfdir_i + \gamma_2 pfdir_i^2 + \gamma_3 ndir_i + \gamma_4 \ln(size)_i + \gamma_5 y10_i + \gamma_6 y12_i + \gamma_7 y14_i + \gamma_8 y16_i + u_{ri}$$

(Model 2_{ROA—nonlinear})

$$tobinsq_i = \beta_0 + \beta_1 pfdir_i + \beta_2 pfdir_i^2 + \beta_3 ndir_i + \beta_4 \ln(size)_i + \beta_5 y10_i + \beta_6 y12_i + \beta_7 y14_i + \beta_8 y16_i + u_{ti}$$

Table 2 Regression analysis of the relationship between profitability and female share in a directorate (Robust standard errors adjusted)

VARIABLES	Model 1 ROA	Model 1 TQ
<i>pfdir</i>	-1.990 (3.175)	0.349 (0.236)
<i>ndir</i>	0.755*** (0.164)	0.0374*** (0.0111)
<i>Insize</i>	-3.132*** (0.345)	-0.436*** (0.0371)
<i>y10</i>	9.769*** (1.731)	0.454*** (0.0883)
<i>y12</i>	9.140*** (1.716)	0.525*** (0.0802)
<i>y14</i>	9.701*** (1.783)	0.767*** (0.0994)
<i>y16</i>	9.366*** (1.793)	0.663*** (0.0979)
<i>Constant</i>	11.61*** (2.916)	2.966*** (0.199)
Observations	840	840
R-squared	0.171	0.296
Adjusted R-squared	.1644699784268874	.2898180175780425

Robust Standard errors in parentheses*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

(Model 2_{TQ—nonlinear})

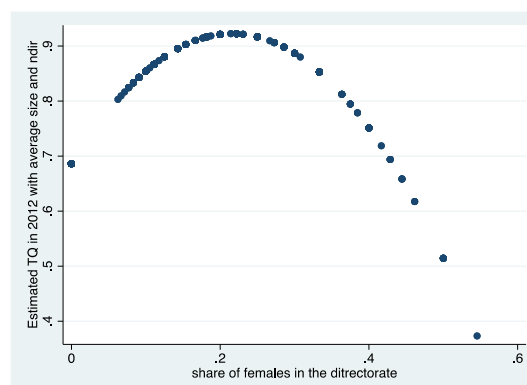
Table 3 Table of Estimation with robust standard errors

VARIABLES	Model 2 ROA_nonlinear	Model 2 TQ_nonlinear
<i>pfdir</i>	6.321 (9.719)	2.188*** (0.602)
<i>pfdir</i> ²	-22.880 (21.434)	-5.063*** (1.365)
<i>ndir</i>	0.720*** (0.176)	0.030*** (0.011)
<i>Insize</i>	-3.140*** (0.345)	-0.438*** (0.037)
<i>y10</i>	9.729*** (1.742)	0.445*** (0.088)
<i>y12</i>	8.995*** (1.759)	0.493*** (0.080)
<i>y14</i>	9.516*** (1.845)	0.726*** (0.099)
<i>y16</i>	9.236*** (1.837)	0.635*** (0.096)
<i>Constant</i>	11.673*** (2.941)	2.979*** (0.198)
Observations	840	840
R-squared	0.172	0.303
Adjusted R-squared	0.165	0.296

Robust Standard errors in parentheses*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Only Model 2_{TQ—nonlinear} is statistically significant.

Introducing a squared term for female share uncovers a significant concave relationship for Tobin's Q: market valuation initially rises with greater female representation, reaches an optimum, and then declines as share increases further. This suggests diminishing marginal returns to diversity beyond an optimal threshold.

**Figure 1** The estimated relationship between profitability and the female share in a directorate for a company observed in 2012

No comparable quadratic effect emerges for ROA, indicating that the benefits of gender diversity are more readily reflected in market assessments than in accounting profitability.

4.3 Interaction effects with Firm Size between profitability and the female share in a directorate

Further analysis with interaction terms between female director proportion and firm size demonstrates a significant moderation effect. Specifically, the impact of gender diversity on Tobin's Q is significantly influenced by firm size.

$$tobinsq_i = \beta_0 + \beta_1 pfdir_i + \beta_2 ndir_i + \beta_3 \ln(size)_i + \beta_4 y10_i + \beta_5 y12_i + \beta_6 y14_i + \beta_7 y16_i + \beta_8 pfdir_i^2 + \beta_9 \ln(size)_i * pfdir_i + \beta_{10} \ln(size)_i * pfdir_i^2 + u_{ti}$$

(Model 3_{TQ interact})

Table 4 The Estimation of Model 3_{TQ interact}

VARIABLES	Model 3 TQ interaction
<i>pfdir</i>	16.104*** (4.142)
<i>ndir</i>	0.034*** (0.012)
<i>lnsize</i>	-0.295*** (0.049)
<i>y10</i>	0.432*** (0.088)
<i>y12</i>	0.466*** (0.081)
<i>y14</i>	0.716*** (0.099)
<i>y16</i>	0.644*** (0.095)
<i>pfdir</i> ²	-36.221*** (8.940)
<i>lnsize</i> * <i>pfdir</i>	-2.271*** (0.614)
<i>lnsize</i> * <i>pfdir</i> ²	5.077*** (1.315)
Constant	2.094*** (0.301)
Observations	840
R-squared	0.319
Adjusted R-squared	0.311

Robust Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

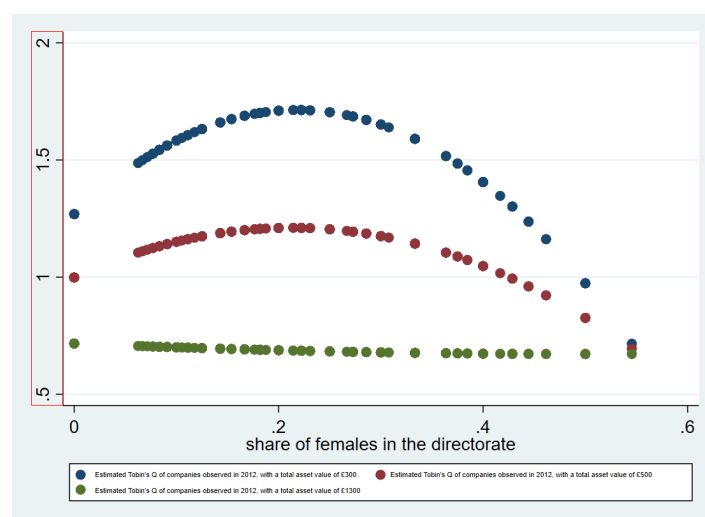


Figure 2 The relationship between profitability and the share of female director regarding the size of the company

Allowing the female share effect to interact with firm size reveals a significantly stronger diversity – valuation link in smaller firms. In these companies, adding female directors' yields pronounced gains in Tobin's Q, whereas in larger firms the same increment contributes less. This finding underscores the importance of organizational context: scale shapes the governance dynamics through which diversity translates into value.

5 Limitation & Future research

Currently, our models include only the female-director share ($pfdir$), its square ($pfdir^2$), board size ($ndir$), firm size ($lnsize$), and two interaction terms ($lnsize*pfdir$ and $lnsize*pfdir^2$). However, we do not account for a range of theoretically relevant variables, which may bias our results. Specifically, we omit human capital (e.g., directors' education level and industry expertise), social capital (e.g., board interlocks and directors' networks), resource dependency (e.g., connections to key external stakeholders), and institutional factors (e.g., legal mandates and governance norms). These dimensions, highlighted by Terjesen, Sealy and Singh (2009), have been shown to influence both board form and firm performance. In addition, other important factors such as board independence, CEO duality and R&D intensity are not included, which may further confound the estimated effects of gender diversity. Future research should incorporate measurable proxies for these constructs to reduce omitted-variable bias and provide a more nuanced understanding of how gender diversity affects firm outcomes (Terjesen et al., 2009).

6 Conclusion

This study investigates the relationship between the share of female directors and firm performance—measured by Return on Assets (ROA) and Tobin's Q—using panel data from United Kingdom listed companies covering 2008–2016.

Our main findings are:

(1) Non-linear effect of gender diversity. We document an inverted-U relationship between the proportion of female directors and both accounting- and market-based performance. Specifically, firm performance increases with female representation up to a threshold (around 20–25 percent), beyond which further increases yield diminishing or even negative returns.

(2) Firm size as a moderator. Larger firms (by logged total assets) benefit more from initial gains in gender diversity but also experience steeper performance declines past the optimal female-director share.

These findings contribute to the board-diversity literature by reconciling prior mixed evidence: gender diversity can be value-enhancing, but only within bounds. The optimal proportion likely reflects a balance between diverse perspectives (information processing benefits) and coordination costs or tokenism pitfalls.

In sum, while female directors can improve firm performance, firms must identify the "sweet spot" of gender balance and foster the board capabilities that turn diversity into a strategic asset.

About the Author

These authors contributed equally to this work and should be considered co-first authors.

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